



Descriptive Statistics- Measures of Central Tendency and variability

1) Provide summary statistics(mean, median, minimum, maximum, standard deviation) for a dataset with numeric variables grouped by one of the qualitative variable.

Importing required libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Importing the dataset

```
sales = pd.read_csv('3.SuperMarket Analysis.csv')
```

sales.head(10)

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity
0	750-67-8428	Alex	Yangon	Member	Female	Health and beauty	74.69	
1	226-31-3081	Giza	Naypyitaw	Normal	Female	Electronic accessories	15.28	
2	631-41-3108	Alex	Yangon	Normal	Female	Home and lifestyle	46.33	
3	123-19-1176	Alex	Yangon	Member	Female	Health and beauty	58.22	
4	373-73-7910	Alex	Yangon	Member	Female	Sports and travel	86.31	
5	699-14-3026	Giza	Naypyitaw	Member	Female	Electronic accessories	85.39	
6	355-53-5943	Alex	Yangon	Member	Female	Electronic accessories	68.84	
7	315-22-5665	Giza	Naypyitaw	Member	Female	Home and lifestyle	73.56	
8	665-32-9167	Alex	Yangon	Member	Female	Health and beauty	36.26	
9	692-92-5582	Cairo	Mandalay	Member	Female	Food and beverages	54.84	

```
sales.tail(10)
```

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quan
990	886-18-2897	Alex	Yangon	Normal	Female	Food and beverages	56.56	
991	602-16-6955	Cairo	Mandalay	Normal	Female	Sports and travel	76.60	
992	745-74-0715	Alex	Yangon	Normal	Male	Electronic accessories	58.03	
993	690-01-6631	Cairo	Mandalay	Normal	Male	Fashion accessories	17.49	
994	652-49-6720	Giza	Naypyitaw	Member	Female	Electronic accessories	60.95	
995	233-67-5758	Giza	Naypyitaw	Normal	Male	Health and beauty	40.35	
996	303-96-2227	Cairo	Mandalay	Normal	Female	Home and lifestyle	97.38	
997	727-02-1313	Alex	Yangon	Member	Male	Food and beverages	31.84	
998	347-56-2442	Alex	Yangon	Normal	Male	Home and lifestyle	65.82	
999	849-09-3807	Alex	Yangon	Member	Female	Fashion accessories	88.34	

```
sales.shape
```

```
(1000, 17)
```

```
# Checking the missing value
```

```
sales.isnull().sum()
```

	0
Invoice ID	0
Branch	0
City	0
Customer type	0
Gender	0
Product line	0
Unit price	0
Quantity	0
Tax 5%	0
Sales	0
Date	0
Time	0
Payment	0
cogs	0
gross margin percentage	0
gross income	0
Rating	0

dtype: int64

```
# Total number of non-missing values--using count()
```

```
sales.count()
```

	0
Invoice ID	1000
Branch	1000
City	1000
Customer type	1000
Gender	1000
Product line	1000
Unit price	1000
Quantity	1000
Tax 5%	1000
Sales	1000
Date	1000
Time	1000
Payment	1000
cogs	1000
gross margin percentage	1000
gross income	1000
Rating	1000

```
dtype: int64
```

```
# What is the total sales amount --using sum()
```

```
sales.sum()
```

0

Invoice ID	750-67-8428226-31-3081631-41-3108123-19-117637...
Branch	AlexGizaAlexAlexAlexGizaAlexGizaAlexCairoCairo...
City	YangonNaypyitawYangonYangonYangonNaypyitawYang...
Customer type	MemberNormalNormalMemberMemberMemberMemberMemb...
Gender	FemaleFemaleFemaleFemaleFemaleFemaleFemaleFema...
Product line	Health and beautyElectronic accessoriesHome an...
Unit price	55672.13
Quantity	5510
Tax 5%	15379.369
Sales	322966.749
Date	01-05-201903-08-201903-03-20191/27/201902-08-2...
Time	1:08:00 PM10:29:00 AM1:23:00 PM8:33:00 PM10:37...
Payment	EwalletCashCredit cardEwalletEwalletEwalletEwa...
cogs	307587.38
gross margin percentage	4761.904762
gross income	15379.369
Rating	6972.7

```
dtype: object
```

```
sales['Sales'].sum()
```

```
np.float64(322966.749)
```

```
sales[['Sales','gross income']].sum()
```

	0
Sales	322966.749
gross income	15379.369

dtype: float64

```
# What is the average satisfaction level and what is the spread of
sales.mean(numeric_only=True)
```

	0
Unit price	55.672130
Quantity	5.510000
Tax 5%	15.379369
Sales	322.966749
cogs	307.587380
gross margin percentage	4.761905
gross income	15.379369
Rating	6.972700

dtype: float64

```
sales['Rating'].mean()
```

```
np.float64(6.9727)
```

```
sales.std(numeric_only=True)
```

```

                                0
Unit price      2.649463e+01
Quantity        2.923431e+00
Tax 5%          1.170883e+01
Sales           2.458853e+02
cogs            2.341765e+02
gross margin percentage  6.131498e-14
gross income    1.170883e+01
Rating          1.718580e+00

```

```
dtype: float64
```

```
sales['Rating'].std()
```

```
1.718580294379123
```

```
# Most frequently ussed payment method--using mode()
```

```
print('The Mode of : ',sales['Payment'].mode)
```

```

The Mode of :  <bound method Series.mode of 0          Ewallet
1             Cash
2      Credit card
3             Ewallet
4             Ewallet
...
995          Ewallet
996          Ewallet
997             Cash
998             Cash
999             Cash
Name: Payment, Length: 1000, dtype: object>

```



```
#Ewallet is the payment method which has been used most frequently
sales['Payment'].mode()
```

```

      Payment
0    Ewallet

dtype: object
```

```
# Minimum and MaximumQuantity sold--using min() and max()
sales['Quantity'].min()    # 1 is the minimum quantity which has bee
1
```

```
sales['Quantity'].max()    # 10 is the maximum quantity which has bee
10
```

```
# Daily increasing sales(how the sales increasing in daily bases)--
sales['Sales'].cumsum().head()    # total amount sold like 548.9715,.
```

```

      Sales
0    548.9715
1    629.1915
2    969.7170
3   1458.7650
4   2093.1435

dtype: float64
```

```
sales['Sales'].head()
```

	Sales
0	548.9715
1	80.2200
2	340.5255
3	489.0480
4	634.3785

dtype: float64

```
#Median
```

```
median = sales['Quantity'].median()
```

```
median
```

```
5.0
```

```
print('median quantity: ' + str(median))
```

```
median quantity: 5.0
```

```
#Var
```

```
var = sales['Quantity'].var()
```

```
var
```

```
8.546446446446451
```

```
print('var of quantity: ' + str(var))
```

```
var of quantity: 8.546446446446451
```

```
#groupby
```

```
groupby_sum = sales.groupby(['City']).sum()
```

```
groupby_sum
```

Invoice ID		Branch									
City											
Mandalay	692-92-5582351-62-0822529-56-3974299-46-180531...	CairoCairoCairoCairoCairoCairoCairoCairoCairoC...									
		Member									
Naypyitaw	226-31-3081699-14-3026315-22-5665300-71-460518...	GizaGizaGizaGizaGizaGizaGizaGizaGizaGizaGizaGi...									
		Normal									
Yangon	750-67-8428631-41-3108123-19-1176373-73-791035...	AlexAlexAlexAlexAlexAlexAlexAlexAlexAlexAlexAl...									
		Membe									

```
groupby_count = sales.groupby(['City']).count()
```

```
groupby_count
```

	Invoice ID	Branch	Customer type	Gender	Product line	Unit price	Quantity
City							
Mandalay	332	332	332	332	332	332	332
Naypyitaw	328	328	328	328	328	328	328
V	310	310	310	310	310	310	310

```
groupby_sum1 = sales.groupby(['Payment']).sum()
```

```
groupby_sum1
```

	Invoice ID		Branch
Payment			
Cash	226-31-3081529-56-3974829-34-3910299-46-180564...	GizaCairoAlexCairoCairoCairoAlexCairoCairoCair...	Naypyitaw
	631-41-3108665-32-9167692-92-5582656-95-934976...	AlexAlexCairoAlexAlexAlexCairoAlexAlexCairoAle...	YangonYang
	750-67-8428123-19-1176373-73-7910699-14-302635...	AlexAlexAlexGizaAlexGizaCairoAlexAlexCairoGiza...	YangonYan

```
groupby_count1 = sales.groupby(['Payment']).count()
```

```
groupby_count1
```

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quan
Payment								
Cash	344	344	344	344	344	344	344	
Credit card	311	311	311	311	311	311	311	
Ewallet	345	345	345	345	345	345	345	

```
print('sum of values, grouped by the payment: ' + str(groupby_sum1))
```

```
sum of values, grouped by the payment:
```

```
Payment
```

```
Cash      226-31-3081529-56-3974829-34-3910299-46-180564...
```

```
Credit card 631-41-3108665-32-9167692-92-5582656-95-934976...
```

```
Ewallet    750-67-8428123-19-1176373-73-7910699-14-302635...
```

```
Branch \
```

```
Payment
```

```
Cash      GizaCairoAlexCairoCairoCairoAlexCairoCairoCair...
```

```
Credit card AlexAlexCairoAlexAlexAlexCairoAlexAlexCairoAle...
```

```
Ewallet    AlexAlexAlexGizaAlexGizaCairoAlexAlexCairoGiza...
```

```
City \
```

```
Payment
```

```
Cash      NaypyitawMandalayYangonMandalayMandalayMandala...
```

```
Credit card YangonYangonMandalayYangonYangonYangonMandalay...
```

```
Ewallet    YangonYangonYangonNaypyitawYangonNaypyitawMand...
```

```
Customer type \
```

```
Payment
```

```
Cash      NormalMemberMemberMemberMemberMemberMemberMemb...
```

```
Credit card NormalMemberMemberMemberMemberMemberMemberMemb...
```

```
Ewallet    MemberMemberMemberMemberMemberMemberMemberMemb...
```

```
Gender \
```

```
Payment
```

```
Cash      FemaleFemaleFemaleFemaleFemaleFemaleFemaleFema...
```

```
Credit card FemaleFemaleFemaleFemaleFemaleFemaleFemaleFema...
```

```
Ewallet    FemaleFemaleFemaleFemaleFemaleFemaleFemaleFema...
```

```
Product line Un
```

Payment			
Cash	Electronic accessories	Electronic accessories	He...
Credit card	Home and lifestyle	Health and beauty	Food and be...
Ewallet	Health and beauty	Health and beauty	Sports and t...

	Quantity	Tax 5%	Sales \
Payment			
Cash	1896	5343.170	112206.570
Credit card	1722	4798.432	100767.072
Ewallet	1892	5237.767	109993.107

		Date \
Payment		
Cash	03-08-2019	03-09-2019
Credit card	03-03-2019	01-10-2019
Ewallet	01-05-2019	1/27/2019

		Time
Payment		
Cash	10:29:00	AM5:03:00
Credit card	1:23:00	PM5:15:00
Ewallet	1:08:00	PM8:33:00

	gross margin percentage	gross income	Rating
Payment			
Cash	1638.095238	5343.170	2397.7
Credit card	1480.952381	4798.432	2178.0

```
print('count of values, grouped by the payment: ' + str(groupby_cou
```

count of values, grouped by the payment:	Invoice ID	Br
Payment		
Cash	344	344
Credit card	311	311
Ewallet	345	345

	Unit price	Quantity	Tax 5%	Sales	Date	Time	cogs
Payment							
Cash	344	344	344	344	344	344	344
Credit card	311	311	311	311	311	311	311
Ewallet	345	345	345	345	345	345	345

	gross margin percentage	gross income	Rating
Payment			
Cash	344	344	344
Credit card	311	311	311
Ewallet	345	345	345

```
#describe
```

```
sales.describe() # it does not provide categorical attribute summr
```

	Unit price	Quantity	Tax 5%	Sales	cogs	perc
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1.000000
mean	55.672130	5.510000	15.379369	322.966749	307.58738	4.761905
std	26.494628	2.923431	11.708825	245.885335	234.17651	6.131413
min	10.080000	1.000000	0.508500	10.678500	10.17000	4.761905
25%	32.875000	3.000000	5.924875	124.422375	118.49750	4.761905
50%	55.230000	5.000000	12.088000	253.848000	241.76000	4.761905
75%	77.935000	8.000000	22.445250	471.350250	448.90500	4.761905

```
sales.describe(include='object')
```

	Invoice ID	Branch	City	Customer type	Gender	Product line	Date	T
count	1000	1000	1000	1000	1000	1000	1000	1
unique	1000	3	3	2	2	6	89	
top	849-09-3807	Alex	Yangon	Member	Female	Fashion accessories	02-07-	7:48


```
sales.describe(include='all')
```

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price
count	1000	1000	1000	1000	1000	1000	1000.000000
unique	1000	3	3	2	2	6	NaN
top	849-09-3807	Alex	Yangon	Member	Female	Fashion accessories	NaN
freq	1	340	340	565	571	178	NaN
mean	NaN	NaN	NaN	NaN	NaN	NaN	55.67213
std	NaN	NaN	NaN	NaN	NaN	NaN	26.49462
min	NaN	NaN	NaN	NaN	NaN	NaN	10.08000
25%	NaN	NaN	NaN	NaN	NaN	NaN	32.87500
50%	NaN	NaN	NaN	NaN	NaN	NaN	55.23000
75%	NaN	NaN	NaN	NaN	NaN	NaN	77.93500
max	NaN	NaN	NaN	NaN	NaN	NaN	99.96000

